

LLM Foundations I: From Natural Language Processing to Transformers

Course Website

- <https://kaychansiri.github.io/research-methods-in-the-age-of-ai/>
- The website is for accessing code used in class, lecture slides, and other supplementary materials.
- **Blackboard is still the main course platform** for checking the class schedule, assignments, announcements, and other course information.

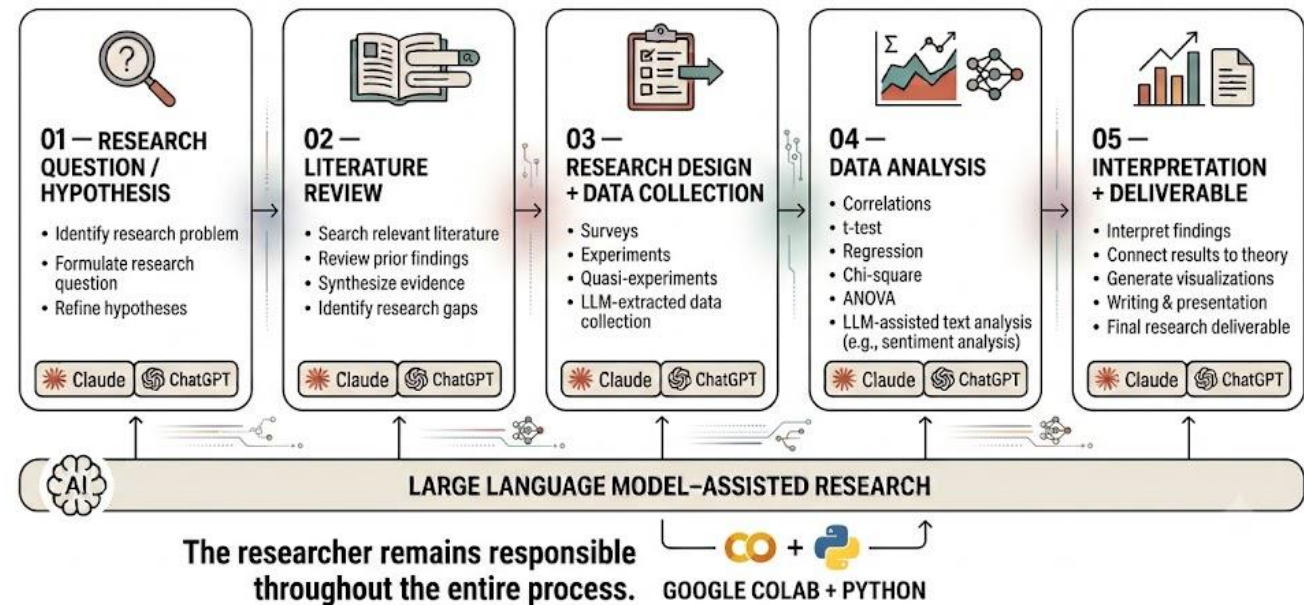
Readings

- Readings will be posted on Blackboard before each course block.

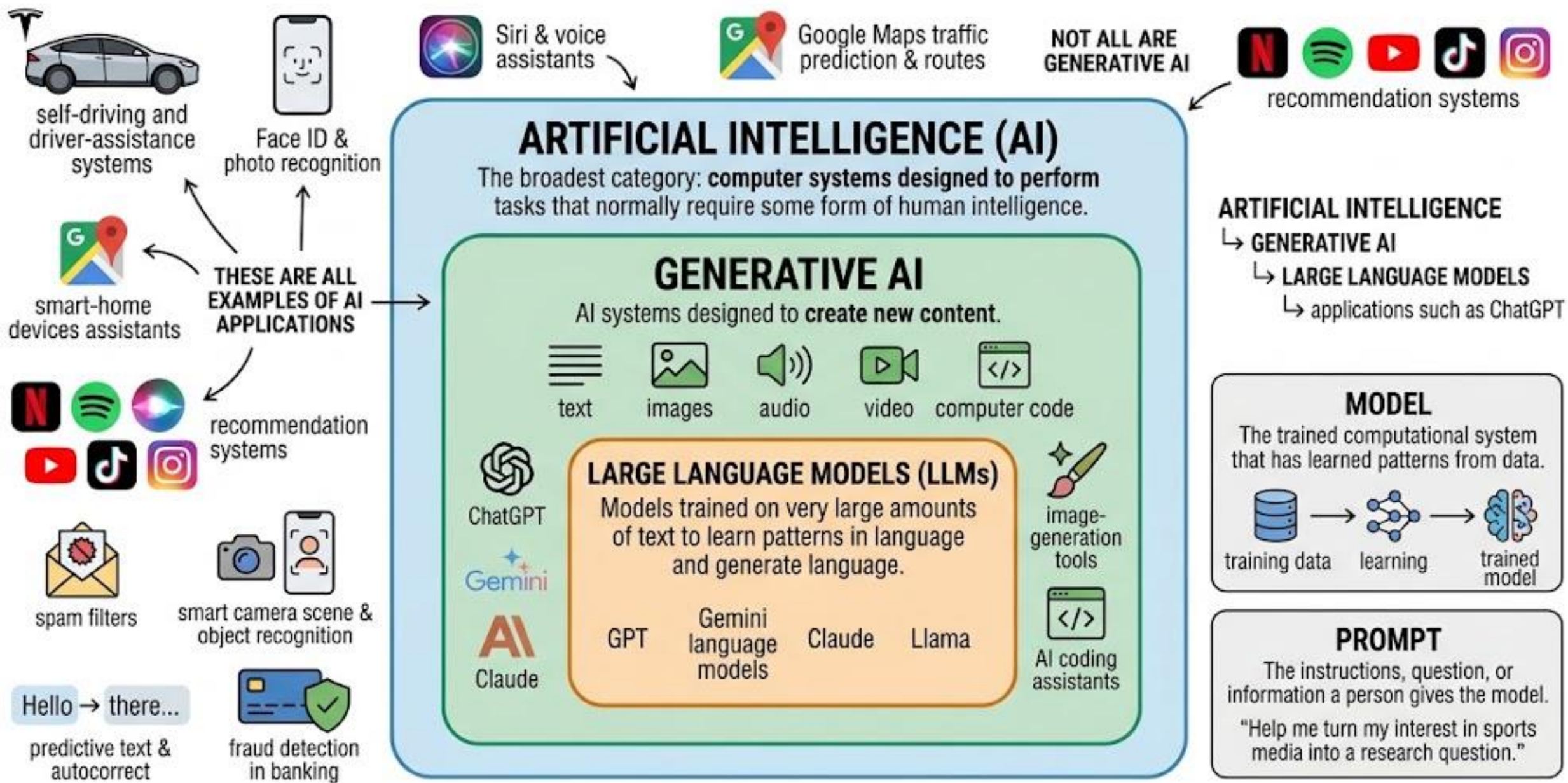
Last Class: What Did We Learn?

- **Research is a process**, not just data analysis.
- **AI can support the entire research process.**
- **LLMs are one type of AI tool** we can use in research.
- **Python + Colab** help us work with data and AI.
- **You are still the researcher** — AI does not replace your judgment.

Research Methods in the Age of AI



What Is AI?



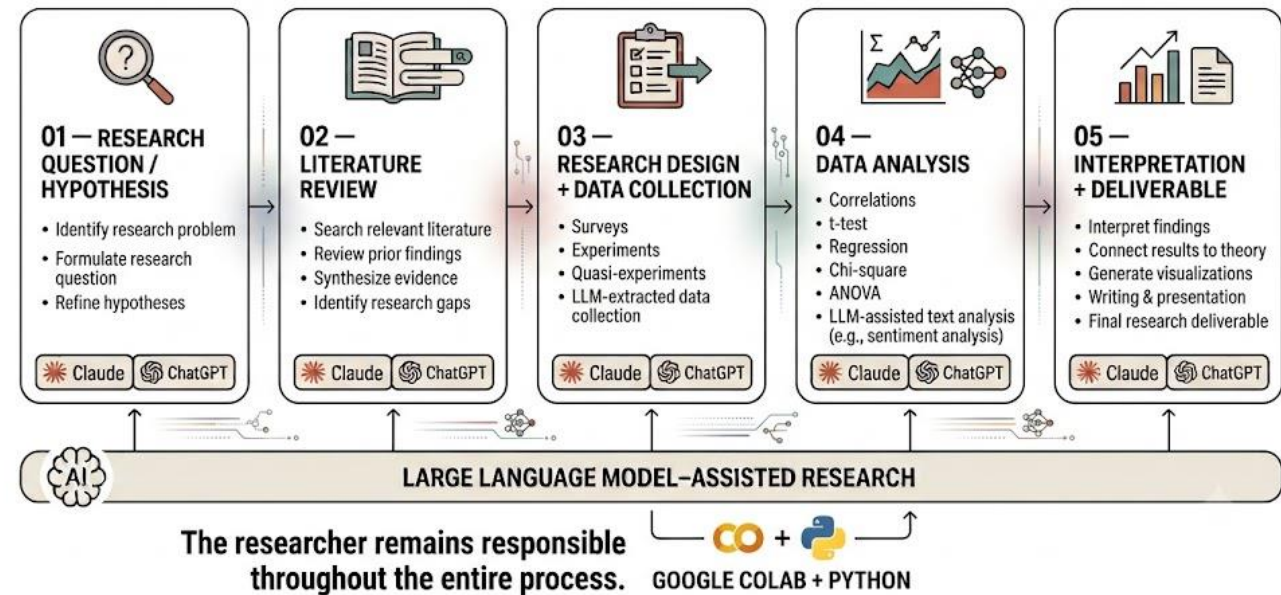
AI ≠ ChatGPT

ChatGPT is one application that allows people to interact with AI models. AI also includes **recommendation systems**, **computer vision**, **autonomous vehicles**, **prediction systems**, **voice assistants**, and many other technologies.

Why Study LLMs?

1. Potential research tool across the process
2. **Knowing the structure reveals the limitations**
3. Can help with text analysis

Research Methods in the Age of AI



Let's Use an LLM to Develop a Research Question

1. Think

What are you curious about?

2. Ask ChatGPT

Help me turn this idea into a research question.

From Natural Language (NLP) Processing to LLMs

What is NLP?

NLP is the part of AI that helps computers **understand and work with human language** in text or speech

'From Rules to Large Language Models: The Evolution of NLP'

1950s–1980s: Rule-Based NLP

'Short Label:
'Humans write explicit
language rules.'



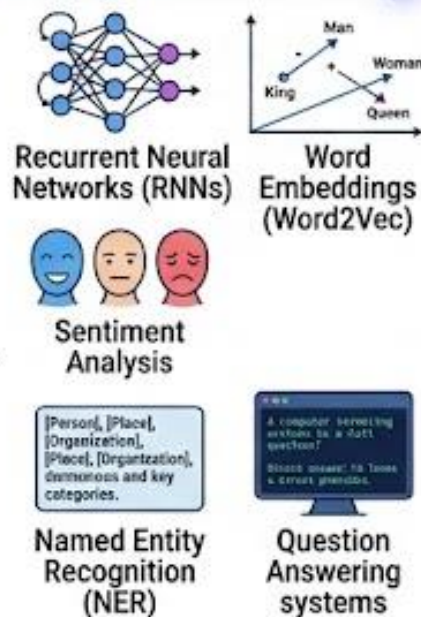
1980s–2000s: Keyword-Based and Statistical NLP

'Computers learn
patterns from word counts
and probabilities.'



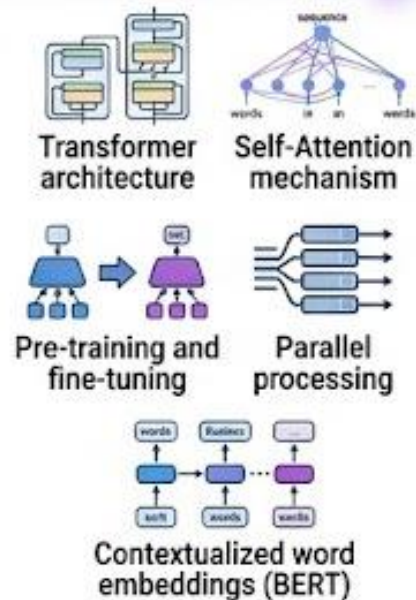
2000s–2016: Machine Learning and Neural NLP

'Artificial neural
networks process
language context.'



2016–2018: The Rise of Transformers

'The 'Transformer'
architecture enables
better sequence learning.'



2018–Present: Large Language Models (LLMs)

'Massive models
pre-trained on vast
internet datasets.'



From NLP to LLMs

	NLP (earlier)	LLMs (current)
Goal	Understand, interpret, generate language	Same goal, much broader scope
Examples	Sentiment analysis, named entity recognition, machine translation	GPT, Claude, LLaMA
Training	Task-specific	Massive, general-purpose
Flexibility	One task per model	Many tasks, minimal retraining
Speed	Slow, process one word at a time, bad at long-range processing	Process all words simultaneously

Vocabulary You'll Need Before We Go Further

Term	Very simple meaning
AI	The broad idea of computers doing tasks that seem “smart”
Generative AI	AI that creates something new, such as text, images, audio, or video
LLM	A type of generative AI that works mainly with language
Model	The trained AI system doing the work
Transformer	The basic technology behind many modern LLMs
Training data	The information the model learns from
Token	A small piece of text the model works with
Model size	Roughly, how big the model is based on how much it has learned internally

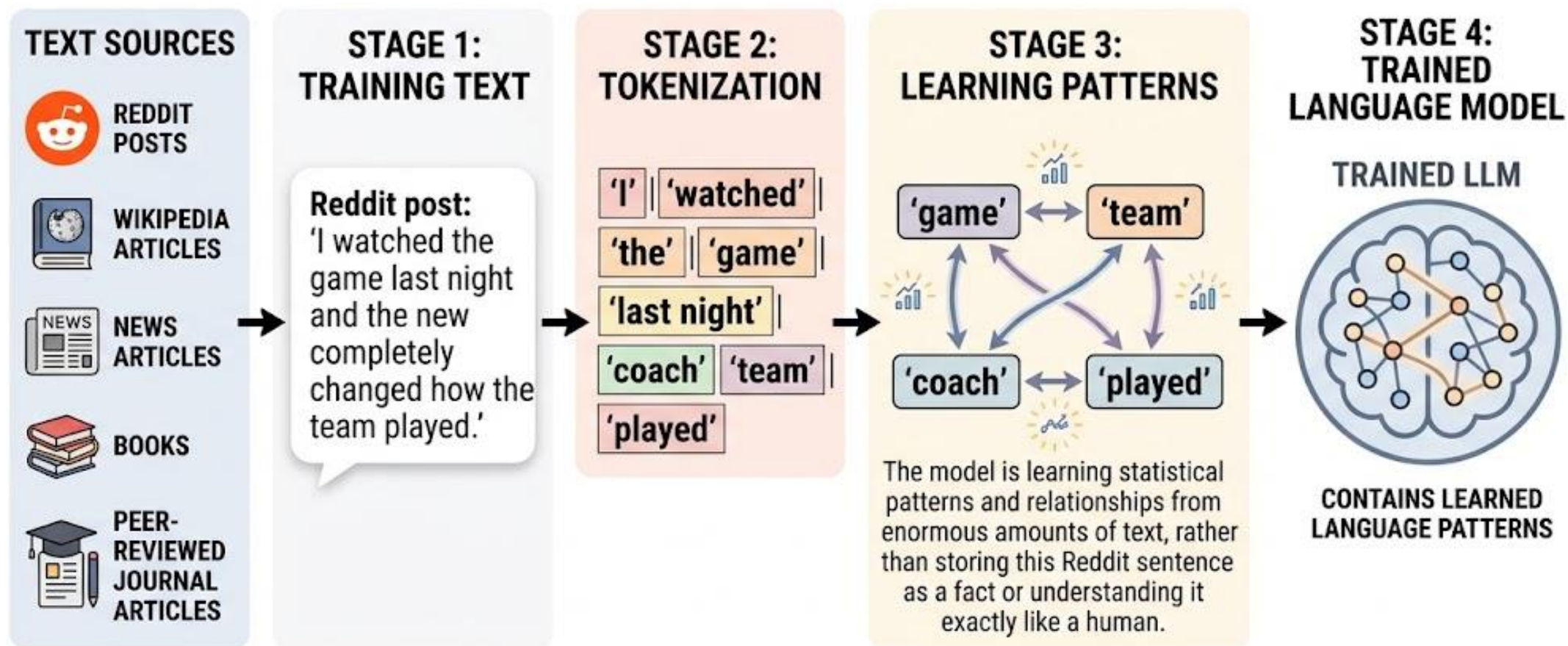
How LLMs Work: Training Time vs. User Time

- **Training time:** the model learns language patterns from large collections of text.
- **User time:** you type a prompt, and the model uses learned patterns to generate a response.
- The model does not “look up” one answer. It predicts likely language based on patterns.

1. Training Time: Learning From Text

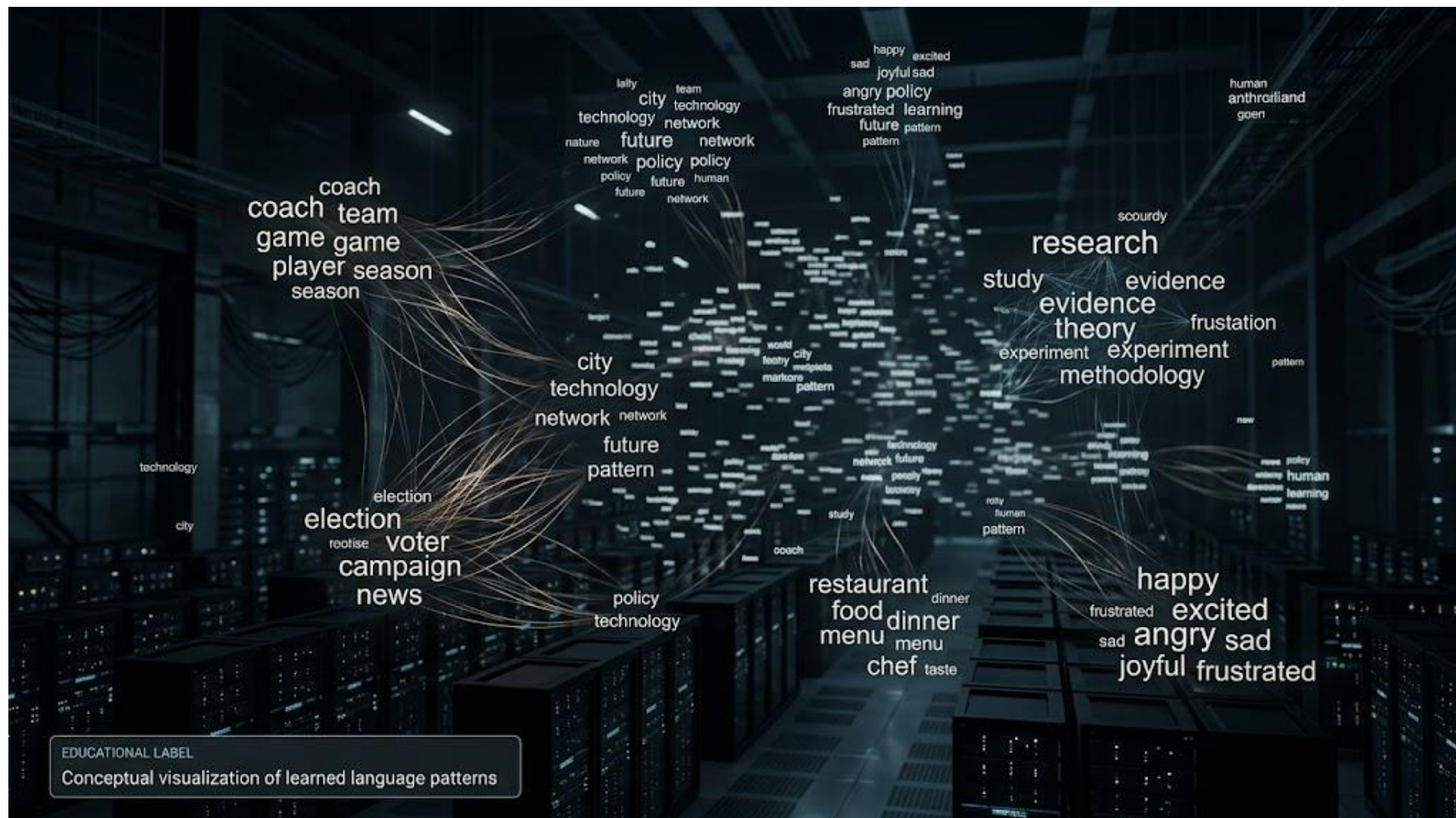
- Text from many sources is broken into smaller pieces called **tokens**.
- The model learns which tokens tend to appear near each other.
- It also learns broader patterns: topics, writing styles, arguments, examples, and associations.
- These patterns are stored as numbers inside the model.

How a Large Language Model (LLM) Learns During Training



Many Text Sources → Tokens → Learned Patterns & Relationships → Trained LLM

Trained LLM



2. User Time: From Prompt to Response

- You enter a prompt.
- The prompt is also broken into tokens.
- The model processes those tokens using the **patterns and relationships it learned during training**.
- It generates a response **one token at a time**.

User Time: From Prompt to Response

STAGE 1 — USER PROMPT

“How does social media use influence sports fans’ trust in teams?”



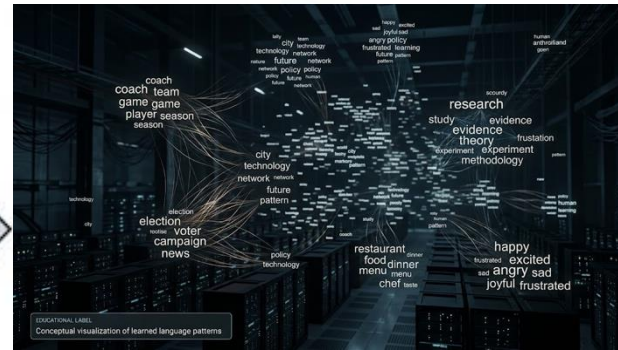
STAGE 2 — TOKENS

“How” “does”
“social media”
“influence” “sports”
“fans” “trust”
“trust” “teams”

“The prompt becomes tokens.”

STAGE 3 — LEARNED PATTERNS

TRAINED LANGUAGE MODEL



TRAINED
LANGUAGE
MODEL

PATTERN
ACTIVATION

“The model processes your prompt using patterns learned during training.”

STAGE 4 — GENERATED RESPONSE

“One possible research question is...”

“One possible research question is... How does social media engagement...”

“One possible research question is... How does social media engagement... relate to sports fans’ trust in professional teams?”

“The model generates a unique response piece by piece.”

PROMPT → TOKENS → LEARNED PATTERNS → GENERATED RESPONSE

Why Different Prompts Give Different Outputs

- Changing words changes the pattern the model follows.
- More specific prompts usually guide the model better.
- Different users may receive different outputs because they provide different context.
- This is why researchers must evaluate the output.

What We Learned

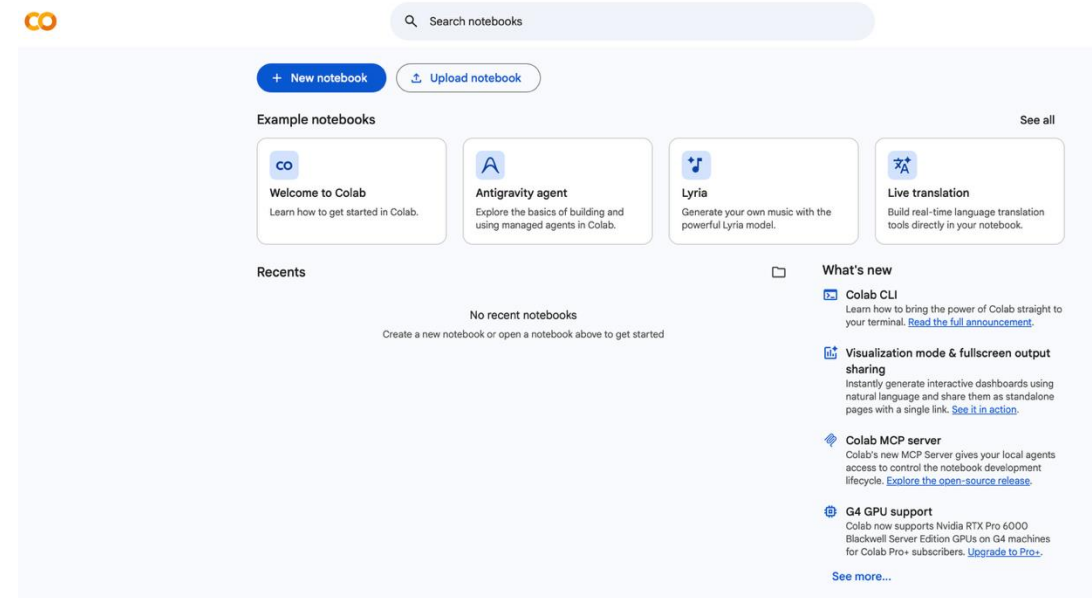
- AI can help turn a broad interest into a researchable question.
- But the human researcher decides whether the question is clear, meaningful, and feasible.
- Better input usually leads to better output.
- AI assistance is useful, but not the same as methodological judgment.

What We Are Not Memorizing Today

- You do **not** need to memorize equations.
- You do **not** need to become an AI engineer.
- You only need to understand the basic logic:
text → tokens → patterns → generated response

Assignment 1 (Due Next Tuesday)

- Go to colab.research.google.com
- Sign in with your Google account
- Click "New Notebook"
- You'll see an empty gray box — that's a code cell
- Take a screenshot of your Colab window
- Upload the screenshot to BlackBoard



Next: Inside the Transformer